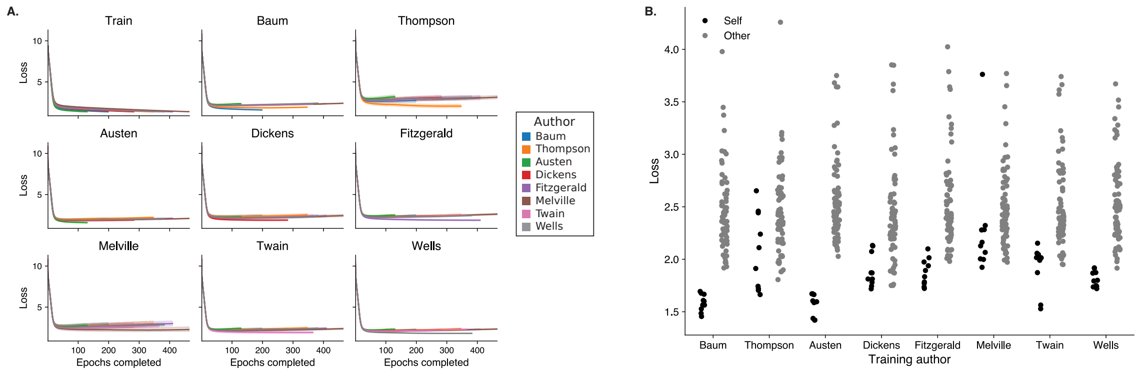
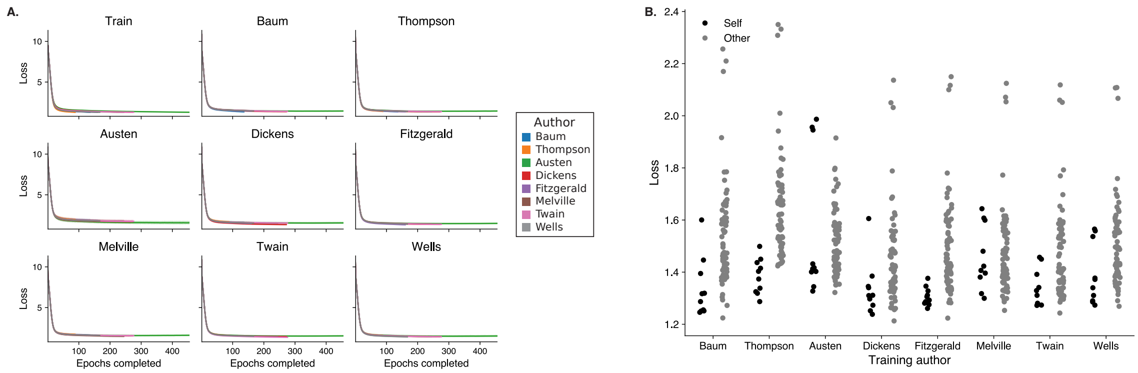
Supplementary materials for: A Stylometric Application of Large Language Models

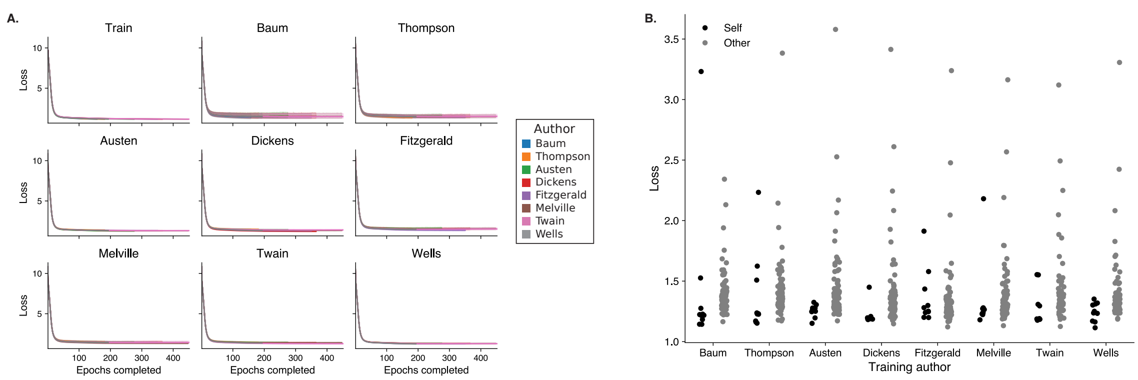
Harrison F. Stropkay, Jiayi Chen, Mohammad J. Latifi,
Daniel N. Rockmore, and Jeremy R. Manning
Dartmouth College
Hanover, NH 03755, USA
{harrison.f.stropkay.25, jiayi.chen.gr, mohammad.javad.latifi.jebelli
daniel.n.rockmore, jeremy.r.manning}@dartmouth.edu



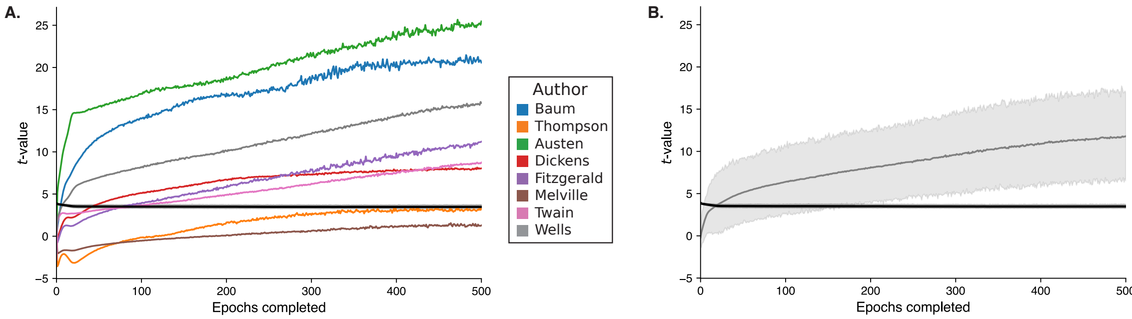
Supplementary Figure 1: Cross-entropy loss across models and authors using only content words. Follows the general format of Figure 1 in the main text, but uses models trained on only content words. All function words are masked out using `<FUNC>`. **A.** Average cross-entropy loss on *Train*ing data and held-out test data from each author, plotted as a function of the number of training epochs. Each color denotes a model trained on a single author's work. Error ribbons denote bootstrap-estimated 95% confidence intervals over 10 random seeds. **B.** Cross-entropy loss assigned to held-out test data by each author's model (x-axis). Held-out test data is either from the *same* author (black) or from *other* authors (gray). Each dot denotes the average loss (across all 1024-token chunks) for a single random seed.



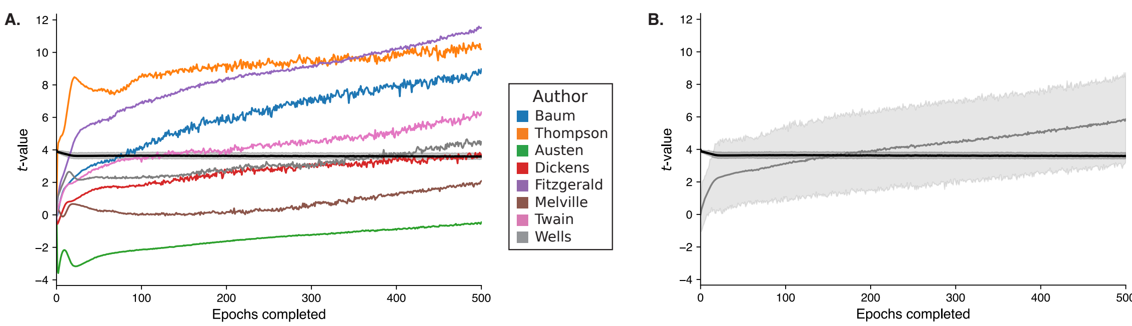
Supplementary Figure 2: Cross-entropy loss across models and authors using only function words. Follows the general format of Figure 1 in the main text, but uses models trained on only function words. All content words are masked out using `<CONTENT>`. **A.** Average cross-entropy loss on *Train*ing data and held-out test data from each author, plotted as a function of the number of training epochs. Each color denotes a model trained on a single author's work. Error ribbons denote bootstrap-estimated 95% confidence intervals over 10 random seeds. **B.** Cross-entropy loss assigned to held-out test data by each author's model (x-axis). Held-out test data is either from the *same* author (black) or from *other* authors (gray). Each dot denotes the average loss (across all 1024-token chunks) for a single random seed.



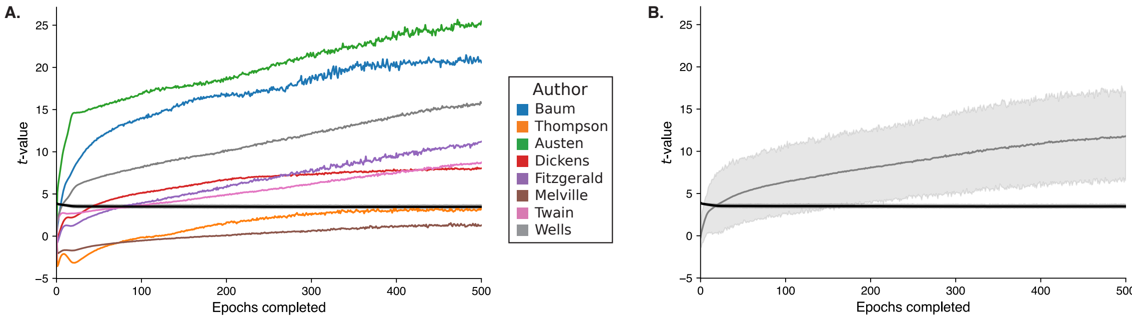
Supplementary Figure 3: Cross-entropy loss across models and authors using only parts of speech. Follows the general format of Figure 1 in the main text, but uses models trained on only parts of speech. All words are replaced with their corresponding part of speech tag. **A.** Average cross-entropy loss on *Train*ing data and held-out test data from each author, plotted as a function of the number of training epochs. Each color denotes a model trained on a single author's work. Error ribbons denote bootstrap-estimated 95% confidence intervals over 10 random seeds. **B.** Cross-entropy loss assigned to held-out test data by each author's model (x-axis). Held-out test data is either from the *same* author (black) or from *other* authors (gray). Each dot denotes the average loss (across all 1024-token chunks) for a single random seed.



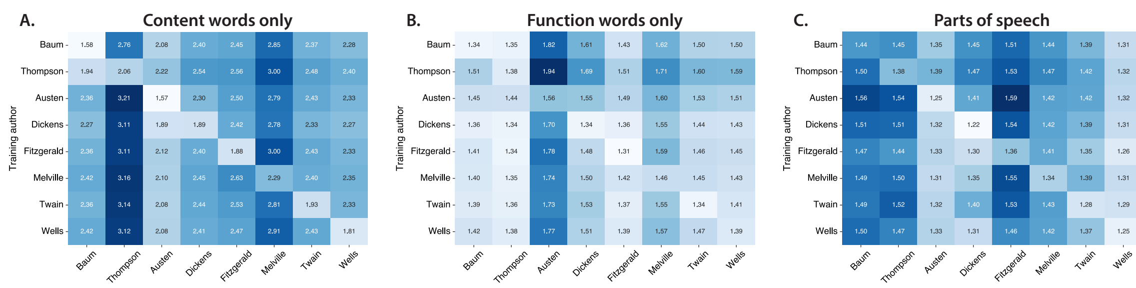
Supplementary Figure 4: Same vs. other author comparisons, by model, using only content words. Follows the general format of Figure 2 in the main text, but uses models trained on only content words. All function words are masked out using `<FUNC>`. **A.** Each curve denotes, as a function of the number of training epochs, the the *t*-statistic from a *t*-test comparing the distribution of losses (across random seeds) assigned to held-out texts from the given author (color) versus held-out texts from all other authors. **B.** The average *t*-statistic across all eight authors, as a function of the number of training epochs. The black curves in both panels indicates the average *t*-value corresponding to $p = 0.001$, for each epoch. Error ribbons denote bootstrap-estimated 95% confidence intervals across authors.



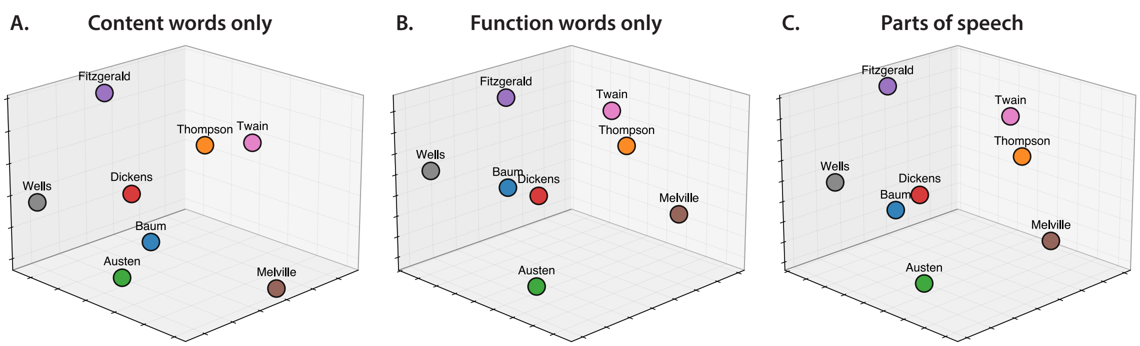
Supplementary Figure 5: Same vs. other author comparisons, by model, using only function words. Follows the general format of Figure 2 in the main text, but uses models trained on only function words. All content words are masked out using `<CONTENT>`. **A.** Each curve denotes, as a function of the number of training epochs, the the *t*-statistic from a *t*-test comparing the distribution of losses (across random seeds) assigned to held-out texts from the given author (color) versus held-out texts from all other authors. **B.** The average *t*-statistic across all eight authors, as a function of the number of training epochs. The black curves in both panels indicates the average *t*-value corresponding to $p = 0.001$, for each epoch. Error ribbons denote bootstrap-estimated 95% confidence intervals across authors.



Supplementary Figure 6: Same vs. other author comparisons, by model, using only parts of speech. Follows the general format of Figure 2 in the main text, but uses models trained on only parts of speech. All words are replaced with their corresponding part of speech tag. **A.** Each curve denotes, as a function of the number of training epochs, the the t-statistic from a t-test comparing the distribution of losses (across random seeds) assigned to held-out texts from the given author (color) versus held-out texts from all other authors. **B.** The average t-statistic across all eight authors, as a function of the number of training epochs. The black curves in both panels indicates the average t-value corresponding to $p = 0.001$, for each epoch. Error ribbons denote bootstrap-estimated 95% confidence intervals across authors.



Supplementary Figure 7: Confusion matrices. Follows the general format of Figure 3 in the main text, but shows confusion matrices for models trained on only content words (A), only function words (B), and only parts of speech (C). Within each panel, the matrix displays the average cross-entropy loss assigned by models trained on each author's writing (column) to held-out texts from each author (row), after subtracting the native author's baseline loss.



Supplementary Figure 8: Multidimensional scaling plots. Follows the general format of Figure 4 in the main text, but shows MDS projections of the (symmetrized) average cross entropy loss matrices shown in Figure 7, for models trained on only content words (A), only function words (B), and only parts of speech (C).

Model	t**-stat**	**df**	p**-value**
Baum	20.58	68.36	5.27×10^{-31}
Thompson	3.29	11.33	6.97×10^{-3}
Austen	25.46	70.33	3.20×10^{-37}
Dickens	8.04	37.39	1.13×10^{-9}
Fitzgerald	11.21	49.02	3.97×10^{-15}
Melville	1.28	10.28	0.2274
Twain	8.73	22.50	1.12×10^{-8}
Wells	15.79	71.87	4.53×10^{-25}

Supplementary Table 1: Loss differences between same-author and other-author texts using only content words. Follows the general format of Table 1 in the main text, but uses models trained on only content words. Each row displays the results of a t-test comparing the average loss values assigned by each author's model (after training is complete) to the author's held-out text and to the other authors' randomly sampled texts.

Model	t**-stat**	**df**	p**-value**
Baum	8.97	20.85	1.34×10^{-8}
Thompson	10.20	22.96	5.39×10^{-10}
Austen	-0.46	9.52	0.6581
Dickens	3.69	17.46	1.73×10^{-3}
Fitzgerald	11.52	77.98	1.70×10^{-18}
Melville	2.08	17.29	0.0529
Twain	6.31	34.66	3.14×10^{-7}
Wells	4.49	15.94	3.76×10^{-4}

Supplementary Table 2: Loss differences between same-author and other-author texts using only function words. Follows the general format of Table 1 in the main text, but uses models trained on only function words. Each row displays the results of a t-test comparing the average loss values assigned by each author's model (after training is complete) to the author's held-out text and to the other authors' randomly sampled texts.

Model	t**-stat**	**df**	p**-value**
Baum	0.04	9.22	0.9695
Thompson	1.05	10.91	0.3179
Austen	5.72	77.97	1.89×10^{-7}
Dickens	4.41	62.85	4.14×10^{-5}
Fitzgerald	0.43	14.25	0.6704
Melville	0.81	11.98	0.4337
Twain	2.43	20.88	0.0240
Wells	4.05	56.90	1.55×10^{-4}

Supplementary Table 3: Loss differences between same-author and other-author texts using only parts of speech. Follows the general format of Table 1 in the main text, but uses models trained on only parts of speech. Each row displays the results of a t-test comparing the average loss values assigned by each author's model (after training is complete) to the author's held-out text and to the other authors' randomly sampled texts.

Embedding-based authorship attribution comparison

To contextualize the performance of our predictive comparison approach, we conducted a comparison using pre-trained text embedding models from the Massive Text Embedding Benchmark (MTEB) leaderboard. We selected three models spanning a range of sizes: `nomic-embed-text-v1.5` (137M parameters), `bge-m3` (568M parameters), and `Qwen3-Embedding-4B` (4.0 billion parameters).

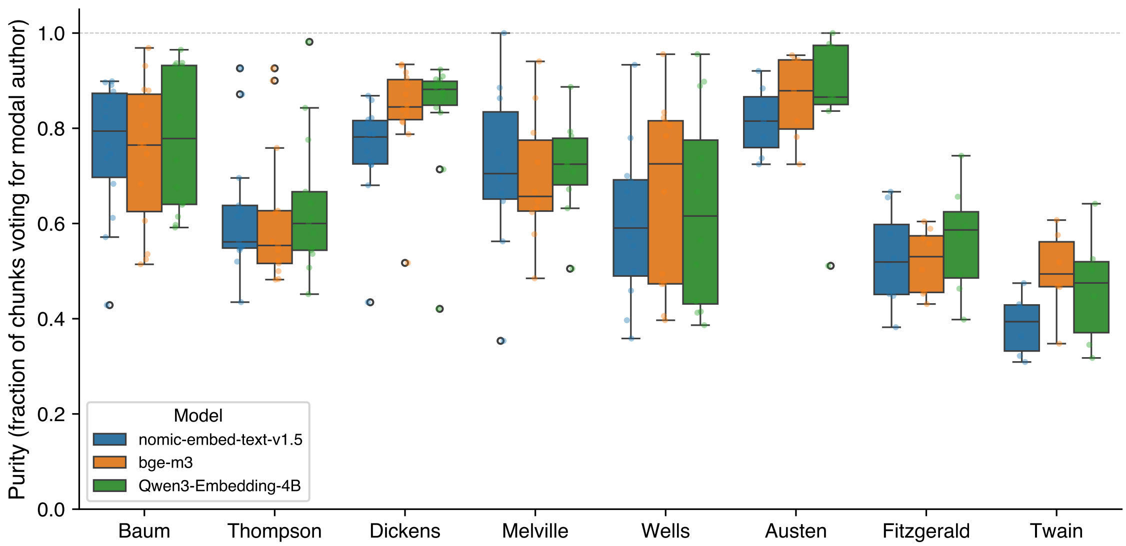
For each model, we chunked each book into 1,024-token windows with 128-token overlap using the GPT-2 tokenizer (yielding 12,164 total chunks across 84 books). For each held-out book, every chunk was classified by its nearest neighbor (cosine similarity) among all chunks from the remaining 83 books. The book-level author prediction was determined by majority vote across chunk-level predictions.

Model	**Parameters**	**Book accuracy**	**Avg. purity**
nomic-embed-text-v1.5	137M	81.0% (68/84)	0.666
bge-m3	568M	76.2% (64/84)	0.694
Qwen3-Embedding-4B	4.0B	70.2% (59/84)	0.703
Predictive comparison (ours)	0.8M $\times$ 8	100% (84/84)	—

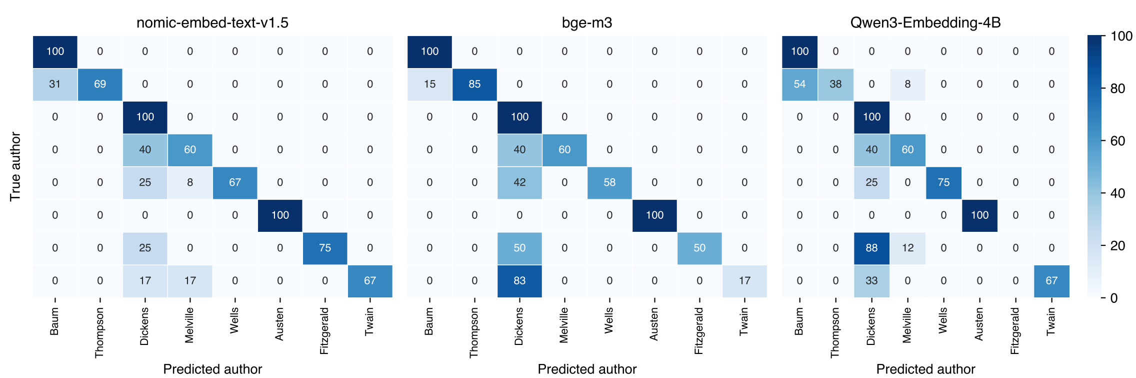
Supplementary Table 4: Book-level attribution accuracy for embedding-based nearest-neighbor classification versus our predictive comparison approach. Purity denotes the fraction of held-out chunks assigned to the modal (predicted) author. Our approach trains one small GPT-2 model per author from scratch; embedding methods use a single pre-trained model for all authors.

We report per-author accuracy and chunk-level purity (the fraction of a held-out book's chunks assigned to the winning author) in Supplementary Figures 9 and 10.

Notable patterns included: (1) authors with the most distinctive styles (Austen, Baum, Dickens) were reliably attributed by all three models at 100% accuracy, whereas authors with fewer books or less distinctive styles (Twain, Melville, Fitzgerald) were frequently misclassified; (2) the largest model (`Qwen3-Embedding-4B`, 4.0B parameters) performed worst overall at 70.2%, completely failing on Fitzgerald (0/8 books correct) and achieving only 38.5% on Thompson—suggesting that larger embedding models may conflate content similarity with stylistic similarity more than smaller ones; (3) the smallest model (`nomic-embed-text-v1.5`, 137M parameters) performed best at 81.0%, indicating that general MTEB ranking does not predict performance on this specialized task; and (4) an inverse relationship between model size and attribution accuracy was observed (81.0%, 76.2%, 70.2% for 137M, 568M, and 4.0B parameters, respectively).



Supplementary Figure 9: Chunk-level purity for embedding-based attribution. Distribution of purity scores (fraction of chunks voting for the modal author) across books, grouped by author and model. Higher purity indicates stronger consensus among chunk-level predictions.



Supplementary Figure 10: Confusion matrices for embedding-based attribution. Each panel shows the percentage of books by each true author (rows) that were classified as each predicted author (columns). Diagonal entries represent correct attributions.